Problem H1: Lorentz-Invariant Differential Cross Section

Mandelstam Variables & the ABC Toy Theory

Particle Physics Study Guide

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1 Problem Statement

Problem Statement

H1) Using the expression for $d\sigma/dt$ for $1+2 \to 3+4$ (natural units throughout):

(a) Express the formula using only Mandelstam variables if $m_1 = m_2 = m$ (and other constants), i.e. replace $|\mathbf{p}_i^*|$ or $|\mathbf{p}_f^*|$ with expressions in s, t, u.

(b) [Somewhat like Griffiths 6.13 or 6.14 but for any frame] Use the result from (a) and the Feynman rules for the ABC theory to derive the cross section $d\sigma/dt$ for $C + C \to A + A$ scattering to lowest order, also assuming propagating particle mass of 0, and the other two particle types have the same non-zero mass m. Write the final expression in terms of only Mandelstam variables s, t, u. Start by writing the diagrams and at least briefly justify why the ABC Feynman rules mean the results wouldn't change essentially.

2 What This Problem Wants: Goals & Objectives

Part (a): What It's Really Asking

Objective: Take the Lorentz-invariant master formula for $d\sigma/dt$ (from Thomson Eq. 3.37) and *eliminate all kinematic variables* ($|\mathbf{p}_i^*|$, $|\mathbf{p}_f^*|$, E^*, θ^*) in favor of the Mandelstam invariants s, t, u and the common mass m.

In plain terms: The starting formula contains the CM momentum p_i^{*2}, which is a frame-dependent-looking quantity. Your job is to use the relation $p_i^{*2} = f(s, m)$ that follows from the definition of s and the equal-mass condition, so the formula becomes manifestly Lorentz-invariant.

Why this matters: The result is a *universal* formula valid in *any* reference frame. You can plug in the matrix element for *any* $2 \to 2$ process (QED, QCD, ABC theory, etc.) and immediately get $d\sigma/dt$ without worrying about what frame you're in.

Part (b): What It's Really Asking

Objective: Apply the result of (a) to a concrete process in the ABC toy theory:

(i) Draw *all* lowest-order Feynman diagrams for $C + C \to A + A$.

(ii) Use the ABC Feynman rules to compute the amplitude $\mathcal{M}$.

(iii) Square it to get $|\mathcal{M}|^2$.

(iv) Plug into the formula from (a) to get $d\sigma/dt$ purely in terms of s, t, u and the coupling g.

Why "the results wouldn't change essentially": The problem notes that Griffiths works out $A + A \to B + B$ in the text (Eqs. 6.54–6.59). For $C + C \to A + A$ the *same* vertex structure (one A, one B, one C at each vertex) applies, so the topology of diagrams is identical—only the labels and masses on the lines change.

3 Background: The Key Ingredients

3.1 The Master Formula (Thomson Eq. 3.37)

Lorentz-Invariant $d\sigma/dt$

For any two-body scattering process $1 + 2 \to 3 + 4$, the Lorentz-invariant differential cross section is:

$$\boxed{\frac{d\sigma}{dt} = \frac{|\mathcal{M}|^2}{64\pi\, s\, p_i^{*2}}} \tag{1}$$

where:

- $s = (p_1 + p_2)^2$ is the CM energy squared,
- $t = (p_1 - p_3)^2$ is the momentum transfer squared,
- p_i^* is the magnitude of either initial-state 3-momentum in the CM frame,
- $|\mathcal{M}|^2$ is the Lorentz-invariant matrix element squared (spin-averaged if applicable).

Derivation sketch: This comes from $d\sigma/d\Omega^* = \frac{p_f^*}{64\pi^2 s\, p_i^*}|\mathcal{M}|^2$ combined with $d\Omega^* = \frac{dt\, d\phi^*}{2p_i^* p_f^*}$ and integration over ϕ^*.

3.2 Mandelstam Variables

Mandelstam Variables for $1 + 2 \to 3 + 4$

$$s = (p_1 + p_2)^2 = (p_3 + p_4)^2 \tag{2}$$
$$t = (p_1 - p_3)^2 = (p_2 - p_4)^2 \tag{3}$$
$$u = (p_1 - p_4)^2 = (p_2 - p_3)^2 \tag{4}$$

with the constraint:

$$\boxed{s + t + u = m_1^2 + m_2^2 + m_3^2 + m_4^2} \tag{5}$$

3.3 CM Momentum in Terms of Masses and s

CM Momentum Formula (Thomson Eq. 3.38)

The magnitude squared of the initial-state CM momentum is:

$$p_i^{*2} = \frac{[s - (m_1 + m_2)^2]\,[s - (m_1 - m_2)^2]}{4s} \tag{6}$$

This is derived by working in the CM frame: $\mathbf{p}_1^* = -\mathbf{p}_2^*$, so $s = (E_1^* + E_2^*)^2$, and solving for $|\mathbf{p}^*|$ using $E_i^{*2} = p^{*2} + m_i^2$.

3.4 ABC Theory Feynman Rules (Griffiths §6.3)

The ABC Toy Theory

Three scalar particles A, B, C (all spin-0, each its own antiparticle) interact via a single primitive vertex connecting one A, one B, one C line.

Feynman Rules:

1. **Vertex factor:** $-ig$ (coupling constant g; in the toy theory, $[g] = \text{mass}$)
2. **Propagator:** $\dfrac{i}{q^2 - m_j^2}$ for internal line of mass m_j carrying 4-momentum q
3. **External lines:** Factor of 1 (scalars have no polarization)
4. **Conservation:** 4-momentum conserved at each vertex
5. **Amplitude:** $-i\mathcal{M} = \prod(\text{vertices}) \times \prod(\text{propagators})$

Why Spin-0 Makes Life Simple

Because all particles in the ABC theory are **scalars**, there are:

- No spin sums or spin averages
- No gamma matrices or trace technology
- No polarization vectors

The amplitude $\mathcal{M}$ is just a *number* (function of Mandelstam variables), not a matrix. This is why the ABC theory is the perfect testing ground for learning the *mechanics* of Feynman diagrams before tackling QED.

4 Solution: Part (a)

Task

Specialize the master formula Eq. (1) to the case $m_1 = m_2 = m$, eliminating p_i^{*2} in favor of s and m.

4.1 Step 1: Simplify p_i^{*2} for Equal Incoming Masses

With $m_1 = m_2 = m$, the general CM momentum formula Eq. (6) becomes:

$$\begin{aligned} p_i^{*2} &= \frac{[s - (m+m)^2]\,[s - (m-m)^2]}{4s} \\ &= \frac{[s - 4m^2]\,[s - 0]}{4s} \\ &= \frac{s(s - 4m^2)}{4s} \end{aligned} \tag{7}$$

CM Momentum for Equal Masses

$$\boxed{p_i^{*2} = \frac{s - 4m^2}{4}} \tag{8}$$

Physical check: At threshold, $\sqrt{s} = 2m$, so $p_i^{*2} = 0$ (particles at rest). For $s \gg 4m^2$, we get $p_i^{*2} \approx s/4$, recovering the ultrarelativistic limit $E^* \approx p^*$.

4.2 Step 2: Substitute into the Master Formula

Inserting Eq. (8) into Eq. (1):

$$\frac{d\sigma}{dt} = \frac{|\mathcal{M}|^2}{64\pi\, s \cdot \dfrac{s - 4m^2}{4}} = \frac{4\,|\mathcal{M}|^2}{64\pi\, s\,(s - 4m^2)} \tag{9}$$

Master Formula in Mandelstam Variables ($m_1 = m_2 = m$)

$$\boxed{\frac{d\sigma}{dt} = \frac{|\mathcal{M}|^2}{16\pi\, s\,(s - 4m^2)}} \tag{10}$$

This is the **universal Lorentz-invariant result** for any $2 \to 2$ scattering with equal-mass incoming particles. It is valid in *any* reference frame—not just the CM frame.

What Makes This "Lorentz-Invariant"?

Every quantity in Eq. (10) is a Lorentz scalar:

- σ is a cross section (Lorentz invariant by construction—it's an "effective area")
- $t = (p_1 - p_3)^2$ is a four-vector scalar product
- $s = (p_1 + p_2)^2$ is a four-vector scalar product
- $|\mathcal{M}|^2$ is built from Lorentz-invariant contractions

So both sides of the equation are frame-independent. This is the power of the Mandelstam formulation: **compute once, use anywhere**.

5 Solution: Part (b) — $C + C \to A + A$ in ABC Theory

Task

Draw the lowest-order Feynman diagrams for $C + C \to A + A$, compute $\mathcal{M}$, and derive $d\sigma/dt$ in terms of s, t, u only.

Mass assignments: $m_B = 0$ (propagating particle), $m_A = m_C = m$ (the "other two particle types").

5.1 Step 1: Identify the Diagrams

Why Only t-channel and u-channel?

In the ABC theory, the *only* vertex connects one A, one B, and one C. For $C + C \to A + A$:

- Each vertex must have exactly one A, one B, one C leg.
- An incoming C converts to an outgoing A by emitting/absorbing a virtual B.
- There is **no s-channel diagram**, because an s-channel would require a $CC \to$ something $\to AA$ vertex, but no single vertex can absorb two C's or emit two A's.

This leaves exactly **two** tree-level diagrams: t-channel and u-channel exchange of B.

Diagram 1 (t-channel):

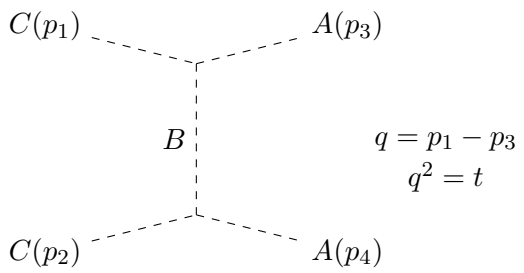
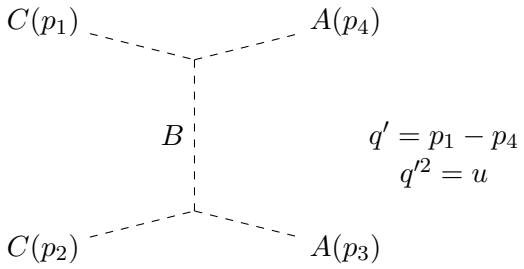


Diagram 2 (u-channel):



Comparison with Griffiths' $A + A \to B + B$

Griffiths works out $A + A \to B + B$ in Eqs. 6.54–6.59. That process also has exactly two diagrams (t-channel and u-channel) with the *same* vertex structure. The only difference is:

- **Griffiths:** external A's, external B's, internal C
- **Our problem:** external C's, external A's, internal B

Since the ABC vertex is *symmetric* (one of each), the combinatorics are identical. The amplitudes differ only by the mass on the internal line: m_C for Griffiths vs. $m_B = 0$ for us. **The structure of the calculation does not change.**

5.2 Step 2: Compute the Amplitude

Applying the Feynman rules to each diagram:

t-channel:

$$-i\mathcal{M}_t = (-ig)^2 \cdot \frac{i}{(p_1 - p_3)^2 - m_B^2} = \frac{-ig^2}{t - m_B^2} \tag{11}$$

u-channel:

$$-i\mathcal{M}_u = (-ig)^2 \cdot \frac{i}{(p_1 - p_4)^2 - m_B^2} = \frac{-ig^2}{u - m_B^2} \tag{12}$$

Total amplitude (sum coherently, since both diagrams contribute to the same process):

$$-i\mathcal{M} = -i\mathcal{M}_t + (-i\mathcal{M}_u) = -ig^2 \left[\frac{1}{t - m_B^2} + \frac{1}{u - m_B^2} \right] \tag{13}$$

Dividing by $-i$:

$$\mathcal{M} = g^2 \left[\frac{1}{t - m_B^2} + \frac{1}{u - m_B^2} \right] \tag{14}$$

Now set $m_B = 0$:

Amplitude for $C + C \to A + A$ (lowest order, $m_B = 0$)

$$\boxed{\mathcal{M} = g^2 \left(\frac{1}{t} + \frac{1}{u} \right) = g^2 \frac{t + u}{tu}} \tag{15}$$

5.3 Step 3: Use the Mandelstam Constraint

With $m_1 = m_2 = m_C = m$ and $m_3 = m_4 = m_A = m$:

$$s + t + u = m_C^2 + m_C^2 + m_A^2 + m_A^2 = 4m^2 \tag{16}$$

$$\implies \quad t + u = 4m^2 - s \tag{17}$$

Substituting into Eq. (15):

Amplitude in Terms of s, t, u

$$\boxed{\mathcal{M} = \frac{g^2(4m^2 - s)}{tu}} \tag{18}$$

5.4 Step 4: Square the Amplitude

Since all particles are spin-0, there is *no* spin averaging:

$$|\mathcal{M}|^2 = \frac{g^4(4m^2 - s)^2}{t^2 u^2} \tag{19}$$

5.5 Step 5: Assemble $d\sigma/dt$

Plug Eq. (19) into the master formula Eq. (10). Note that the incoming particles are C, C with $m_C = m$, so we use $m_1 = m_2 = m$ directly.

Additionally, since the final state contains **two identical particles** (A and A, where A is its own antiparticle), we include the statistical factor $S = \frac{1}{2}$:

$$\frac{d\sigma}{dt} = \frac{S \cdot |\mathcal{M}|^2}{16\pi\, s\,(s-4m^2)} = \frac{1}{2} \cdot \frac{g^4(4m^2-s)^2}{16\pi\, s\,(s-4m^2)\,t^2u^2} \tag{20}$$

Now use $(4m^2 - s)^2 = (s - 4m^2)^2$:

$$\frac{d\sigma}{dt} = \frac{g^4(s-4m^2)^2}{32\pi\, s\,(s-4m^2)\,t^2u^2} = \frac{g^4(s-4m^2)}{32\pi\, s\,t^2u^2} \tag{21}$$

Final Result: $d\sigma/dt$ for $C + C \to A + A$

$$\boxed{\frac{d\sigma}{dt} = \frac{g^4\,(s-4m^2)}{32\pi\, s\,t^2\,u^2}} \tag{22}$$

where $s + t + u = 4m^2$, $m_A = m_C = m$, $m_B = 0$, and the factor of $\frac{1}{2}$ for identical final-state particles has been included.

Physical Checks on the Result

1. **Threshold behavior:** As $s \to 4m^2$, $d\sigma/dt \to 0$. This is correct: at threshold the particles have zero CM momentum and can't scatter.
2. **Forward/backward divergence:** The $1/t^2$ and $1/u^2$ factors cause $d\sigma/dt \to \infty$ as $t \to 0$ or $u \to 0$ (i.e., $\theta^* \to 0$ or $\theta^* \to \pi$). This is the hallmark of massless exchange—analogous to Rutherford/Coulomb scattering from a $1/r$ potential. If $m_B \neq 0$, these divergences would be regulated.
3. **Dimensions:** $[g^4] = \text{mass}^4$, $[s\,t^2\,u^2] = \text{mass}^{10}$, $(s - 4m^2)$ has dimensions mass^2. So $d\sigma/dt$ has dimensions mass^{-4}, consistent with [area/mass^2] in natural units. ✓
4. **Symmetry:** The final state has two identical A's, so $d\sigma/dt$ should be symmetric under $3 \leftrightarrow 4$, which is $t \leftrightarrow u$. The formula has $t^2u^2 = (tu)^2$, which is indeed symmetric. ✓

6 Without the Statistical Factor

Some conventions (including parts of Griffiths and Thomson) do *not* include the statistical factor S in the differential cross section formula, absorbing it instead into the total cross section via limits of integration. If your course omits S:

Alternative (No Statistical Factor)

$$\left.\frac{d\sigma}{dt}\right|_{\text{no } S} = \frac{g^4\,(s-4m^2)}{16\pi\, s\,t^2\,u^2} \tag{23}$$

Check with your instructor which convention is expected. The difference only matters when computing the total cross section by integrating over t: with S, you integrate over the full range; without S, you integrate over half the range to avoid double-counting.

7 Summary of Key Formulas

Quantity	**Expression**
CM momentum ($m_1 = m_2 = m$)	$p_i^{*2} = \dfrac{s - 4m^2}{4}$
Master formula	$\dfrac{d\sigma}{dt} = \dfrac{\lvert\mathcal{M}\rvert^2}{16\pi\, s\,(s - 4m^2)}$
Mandelstam constraint	$s + t + u = 4m^2$
$C + C \to A + A$ amplitude	$\mathcal{M} = \dfrac{g^2(4m^2 - s)}{tu}$
$d\sigma/dt$ (with $S = 1/2$)	$\dfrac{g^4(s - 4m^2)}{32\pi\, s\, t^2\, u^2}$

8 Practice Questions for Self-Study

1. **Derive** $p_i^{*2} = \frac{[s-(m_1+m_2)^2][s-(m_1-m_2)^2]}{4s}$ from scratch, starting in the CM frame with $E_1^{*2} = p^{*2} + m_1^2$ and $\sqrt{s} = E_1^* + E_2^*$.

2. **Explain** why there is no s-channel diagram for $C + C \to A + A$ in the ABC theory. What process *would* have an s-channel contribution?

3. **Repeat** the calculation for $A + A \to B + B$ (Griffiths' original example) with $m_A = m$, $m_B = 0$, and internal particle C of mass m. Show that:

$$\frac{d\sigma}{dt} = \frac{g^4(t + u)^2}{32\pi\, s\,(s - 4m^2)(t - m^2)^2(u - m^2)^2}$$

4. **Show** that in the CM frame with $m_A = m_C = m$ and $m_B = 0$:

$$t = -\frac{s - 4m^2}{2}(1 - \cos\theta^*), \qquad u = -\frac{s - 4m^2}{2}(1 + \cos\theta^*)$$

   and verify that $s + t + u = 4m^2$.

5. **Verify the dimensions** of $d\sigma/dt$ in natural units, confirming it has dimensions of GeV^{-4}.